

Sang Min Lee

Ph.D. Candidate in Artificial Intelligence · Structural & Wind Engineering
Seoul, Republic of Korea

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Summary

- Ph.D. Candidate in Artificial Intelligence, Seoul National University (Advisor: Prof. Thomas Kang)
- Deep learning for structural and wind engineering: wind-resistant design, structural health monitoring, and non-destructive evaluation of concrete
- Data-driven methods (deep learning, representation learning, large language models) for wind force reconstruction, automated impact-echo testing, and concrete crack detection

Research Interests

My research interests include, but are not limited to:

- Deep learning for wind-resistant design of buildings (wind load, pressure, and force estimation and reconstruction)
- Machine learning for structural health monitoring and anomaly detection
- Non-destructive evaluation of concrete structures (automated impact-echo testing, crack detection)
- Data-driven prediction of structural and material behavior
- Large language models for structural design codes (retrieval-augmented generation, mixture-of-experts)
- Physics-informed and data-driven modeling for civil and architectural engineering

Education

Seoul National University

Seoul, Republic of Korea

Ph.D. in Artificial Intelligence (Interdisciplinary Program)

Mar. 2023 - Feb. 2027 (expected)

- Dissertation: “Deep Learning Application for Wind-Resistant Design of Building Structure”
- Advisor: Prof. Thomas Kang

Seoul National University

Seoul, Republic of Korea

M.S. in Architecture and Architectural Engineering

Mar. 2021 - Feb. 2023

- Thesis: “Machine Learning Application to Automated Impact-Echo Test for Concrete Structures”
- Advisor: Prof. Thomas Kang

Seoul National University

Seoul, Republic of Korea

B.S. in Electrical and Computer Engineering (Double Major)

Sep. 2017 - Feb. 2021

Seoul National University

Seoul, Republic of Korea

B.S. in Architecture and Architectural Engineering

Mar. 2016 - Feb. 2021

Korea Science Academy of KAIST

Busan, Republic of Korea

Science High School for the Gifted

Mar. 2013 - Feb. 2016

Experience

Structural Engineering Laboratory, Seoul National University

Seoul, Republic of Korea

Graduate Research Assistant

Mar. 2021 - Present

- Research projects: Hyper-converged Forensic Research Center for Infrastructure (2021–2027, NRF/MSIT); Smart City Innovative Technology Demonstration (2024–2025, KAIA/MOLIT); Outstanding Patent Prototype Development (2023, SNU); AI for Construction (2024, SNU/Sejong Construction)

Wind and Structures (peer-reviewed SCIE journal)

Academic Reviewer

Jan. 2024 - Present

- Volunteer reviewer in the field of wind engineering

The University of Oklahoma

Norman, OK, USA

Visiting Student Researcher

Oct. 2025 - Nov. 2025

- School of Civil Engineering and Environmental Sciences & National Weather Center

Wind Engineering Institute of Korea

Seoul, Republic of Korea

Conference Organizing Committee Member

Apr. 2025 - May 2025

- Organizing committee, 28th WEIK Annual Conference & General Assembly (hosted at SNU)

- Architecture and AI (22S, 23S, 24S): guest lecturer on hands-on deep neural networks
- Introduction to IoT, AI, and Big Data (21S, 21F, 22S, 22F): IoT project evaluator; Arduino UNO; head TA (22F)
- Creative Engineering Design (21F, 22S): IoT project evaluator; Arduino UNO, ESP32
- IoT and Creative Engineering Design (21F, 22S): IoT project evaluator; Arduino UNO, Teensy, ESP32, STM boards
- Structural Experiment and Material (22S): Euler buckling, truss deflection, shake-table seismic test
- Mechanics of Materials for Architectural Engineering (21F): fundamentals for structural engineering
- Prestressed Concrete (21F): class management for post-tensioned structures
- Computation in Architectural Engineering (21S): MATLAB for structural engineering students

- HpSE, SNU (2021): wind pressure coefficient prediction for wind-resistant building design
- SNUCEM (2019–2020): AHP analysis for improvement of military dining facilities
- AEPRL, SNU (2018–2019): IoT device (hardware and software) for indoor surface-condensation detection

- Data analysis for building energy performance evaluation (ECO2, AutoCAD, Python)

Publications

International Journal Articles

1. Lee, S. M., Hong, J., Choi, H., and Kang, T. H.-K. (2025) "Machine learning assisted method for automated impact-echo testing of concrete structure," *Journal of Nondestructive Evaluation*, 44(4), 121.
2. Lee, S. M., Choi, H.-S., Kim, C., and Kang, T. H.-K. (2025) "Lightweight alternative machine learning model for automating concrete crack image classification," *ACI Structural Journal*, 122(5).
3. Lee, Y. I., Lee, S. M., and Kang, T. H.-K. (2025) "Multiclass deep support vector data description for structural health monitoring," *ASCE Journal of Structural Engineering*, 151(12).
4. Lee, S. M., Alinejad, H., and Kang, T. H.-K. (2026) "Wind force time history reconstruction with reduced pressure taps using deep learning," *Engineering Applications in Artificial Intelligence*. Under review.
5. Yan, S., Gupta, H., Chen, M., Zhu, S., Li, Z., Wen, Y., Zhang, M., Cao, J., Chen, X., Lee, S. M., Kang, T. H.-K., Duan, Q., Sorooshian, S., and Hong, Y. (2026) "Towards a reasoning-based physics-informed framework for efficient parameter specification in earth system modeling," *Science Advances*. Under review.
6. Lee, S. M., Yan, S., Hong, Y., and Kang, T. H.-K. (2026) "Synoptic and non-synoptic wind classification using pseudo-labeling and machine learning with the Oklahoma Mesonet database," *Journal of Wind Engineering and Industrial Aerodynamics*. Under review.

Domestic Journal Articles

1. Ahn, B., Lee, D., Lee, S. M., and Kang, T. H.-K. (2023) "Research on satellite image-based wind speed prediction methodology using neural network," *Journal of Wind Engineering Institute of Korea*, 27(4), 135–142. [In Korean]
2. Lee, S. M., Sung, H. S., and Kang, T. H.-K. (2022) "Comparison of performance for predicting compressive strength of concrete using machine learning," *Journal of the Korea Concrete Institute*, 34(5), 505–513. [In Korean]

International Conference Proceedings

1. Lee, S. M. and Kang, T. (2025) "Retrieval augmented generation for assistance of using wind load design codes," *The 10th Asia-Pacific Conference on Wind Engineering (APCWE10)*, Chengdu, China.
2. Lee, S. M., Lee, Y. I., and Kang, T. (2024) "Multi-sphere deep SVDD for system identification of building structure," *The 2024 World Congress on Advances in Civil, Environmental, & Materials Research*, Seoul, Republic of Korea.
3. Lee, S. M., Choi, H.-S., Kim, C., and Kang, T. (2024) "Representation learning for classification of concrete crack images," *The 2024 Structures Congress*, Seoul, Republic of Korea.
4. Lee, S. M. and Kang, T. (2022) "Clustering wind pressure tap using dynamic time warping," *8th European-African Conference on Wind Engineering (EACWE2022)*, Bucharest, Romania.
5. Lee, S. M. and Kang, T. (2022) "Shear strength prediction of prestressed concrete based on machine learning," *The 2022 Structures Congress*, Seoul, Republic of Korea.
6. Lee, S. M. and Kang, T. (2021) "Prediction of wind pressure coefficients on high-rise building façade using LSTM RNN model for sensor reduction," *The 2021 World Congress on Advances in Structural Engineering and Mechanics*, Seoul, Republic of Korea.

Domestic Conference Proceedings

1. **Lee, S. M.** and Kang, T. (2026) “Deep learning-based reconstruction of wind force time history from reduced pressure tap measurements,” *The Wind Engineering Institute of Korea 2026 Conference*, Gwangju, Republic of Korea. [In Korean]
2. **Lee, S. M.** and Kang, T. (2026) “Development of edge AI impact echo system for internal defect inspection in concrete structure,” *Korea Concrete Institute 2026 Spring Conference*, Changwon, Republic of Korea. [In Korean]
3. Park, H. G., **Lee, S. M.**, and Kang, T. (2026) “Application of impact-echo testing to liner-plate-backed concrete with simulated internal defects in nuclear containment buildings,” *Korea Concrete Institute 2026 Spring Conference*, Changwon, Republic of Korea. [In Korean]
4. **Lee, S. M.** and Kang, T. (2025) “Clustering of wind pressure coefficients on concrete building facades using deep learning,” *Korea Concrete Institute 2025 Spring Conference*, Jeju, Republic of Korea. [In Korean]
5. Lee, Y. I., **Lee, S. M.**, and Kang, T. (2024) “Structural health monitoring of concrete building using multi-class deep SVDD,” *Korea Concrete Institute 2024 Fall Conference*, Sokcho, Republic of Korea. [In Korean]
6. **Lee, S. M.** and Kang, T. (2024) “Framework of large language model for structural design code using mixture of experts,” *Korea Concrete Institute 2024 Fall Conference*, Sokcho, Republic of Korea. [In Korean]
7. **Lee, S. M.**, Choi, H.-S., Kim, C., and Kang, T. (2024) “Lightweighting concrete crack image classification machine learning models for on-device AI,” *Korea Concrete Institute 2024 Spring Conference*, Gwangju, Republic of Korea. [In Korean]
8. **Lee, S. M.**, Choi, H., and Kang, T. (2023) “Automation of impact-echo test using machine learning,” *Korea Concrete Institute 2023 Spring Conference*, Busan, Republic of Korea. [In Korean]
9. **Lee, S. M.**, Sung, H. S., and Kang, T. (2022) “Prediction of concrete compressive strength using machine learning,” *Korea Concrete Institute 2022 Spring Conference*, Jeju, Republic of Korea. [In Korean]
10. **Lee, S. M.** and Kang, T. (2021) “Prediction of wind pressure coefficient for concrete building wall using LSTM RNN,” *Korea Concrete Institute 2021 Fall Conference*, Gyeongju, Republic of Korea. [In Korean]
11. **Lee, S. M.** and Kang, T. (2021) “Prediction of wind pressure coefficients for concrete building walls using LSTM network,” *Korea Concrete Institute 2021 Spring Conference*, Yeosu, Republic of Korea. [In Korean]

Patents

1. **Lee, S. M.** and Kang, T. (2022) “Wind load estimation system based on artificial intelligence,” KR 10-2611457 (registered) & PCT/KR2023/021 (applied).
2. **Lee, S. M.** and Kang, T. (2023) “AI-based defect detection system inside concrete members,” KR 10-2480382 (registered) & PCT/KR2023/0063 (applied).
3. **Lee, S. M.** and Kang, T. (2024) “AI-based defect detection system inside concrete members,” US 18/875,721 (applied).
4. **Lee, S. M.** and Kang, T. (2025) “AI-based defect detection system inside concrete members,” EP 23915029.5 (applied).

Honors & Awards

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| 1. AI Fellowship , Graduate School of Artificial Intelligence, SNU | 2025 & 2026 |
| 2. AI Paper Competition Excellence Award , Graduate School of Artificial Intelligence, SNU | 2025 |
| 3. Graduate School of Artificial Intelligence Fellowship , Seoul National University | 2023 & 2024 |
| 4. Best Teaching Assistant Award , College of Engineering, SNU | 2022 |
| 5. Conference Outstanding Paper Award , Korea Concrete Institute | 2022 |
| 6. Full Scholarship (SEN Engineering Group) , Dept. of Architecture & Architectural Eng., SNU | 2021 |

Licensure & Certification

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| 1. Big Data Analysis Engineer (National Technical Qualification) , Korea Data Agency | 2025 |
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Professional Membership

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| 1. Student Member , American Society of Civil Engineers (ASCE) | 2025– |
| 2. Student Member , Wind Engineering Institute of Korea | 2023– |
| 3. Student Member , Korea Institute of Structural Maintenance and Inspection | 2023– |
| 4. Student Member , Architectural Institute of Korea | 2023– |
| 5. Student Member , American Concrete Institute (ACI) | 2021– |
| 6. Student Member , Korea Concrete Institute | 2021– |